Rollno :241001307

IMPLEMENTING A* ALGORITHM

Python Code:

```
from queue import PriorityQueue



def a_star(graph, start, goal, heuristic):

    open_set = PriorityQueue()
    open_set.put((0, start))

    came_from = {}

    g_score = {node: float('inf') for node in graph}
    g_score[start] = 0

    while not open_set.empty():

        _, current = open_set.get()

        if current == goal:
```

```
            path = []

            while current in came_from:
                path.append(current)
                current = came_from[current]

            path.append(start)

            return path[::-1]

        for neighbor, cost in graph[current]:

            tentative_g = g_score[current] + cost

            if tentative_g < g_score[neighbor]:

                came_from[neighbor] = current
                g_score[neighbor] = tentative_g

                f = tentative_g + heuristic[neighbor]

                open_set.put((f, neighbor))

    return None
```

```python
graph = {
    'A': [('B', 1), ('C', 4)],
    'B': [('D', 2), ('E', 5)],
    'C': [('F', 1)],
    'D': [('G', 2)],
    'E': [('G', 1)],
    'F': [('G', 3)],
    'G': []
}

heuristic = {
    'A': 7,
    'B': 6,
    'C': 2,
    'D': 1,
    'E': 0,
    'F': 3,
    'G': 0
}

path = a_star(graph, 'A', 'G', heuristic)

print("Shortest path using A*:", path)
```

Output:

Shortest path using A*: ['A', 'B', 'D', 'G']

Result:

Thus the program of using the MINIMAX algorithm to evaluate the optimal move in a game tree was executed successfully.